Shyam Venkatasubramanian

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Personal website

Education

2021–2025 **Ph.D. in Electrical and Computer Engineering**, *Duke University*.
Advised by Vahid Tarokh.

2018–2021 **B.S. in Electrical Engineering**, *UCLA*.
Magna Cum Laude

Professional Experience

2025–Present **Head of Technical Staff**, *Anthrogen*.

Spring 2024 **Machine Learning Intern**, *Tesla*.
Autopilot AI and navigation team. Supervised by Dariush Dabiri.

Summer 2023 **Research Intern**, *U.S. Air Force Research Laboratory (AFRL)*.
Joint project with SPAS (Duke). Supervised by Muralidhar Rangaswamy.

Summer 2019 **Engineering Intern**, *Schweitzer Engineering Laboratories (SEL)*.
Supervised by Marcos Donolo.

Summer 2018 **Technical Assistant**, *Washington State University EECS Department*.
Supervised by Anjan Bose.

Research Experience

2021–2025 **Duke Signal Processing and Applied Statistics Group (SPAS)**.
Thesis: *Advancing Deep Learning for Data and Signal Processing Applications*.

2020–2021 **UCLA Laboratory for Robust Information Systems (LORIS)**.
Supervised by Lara Dolecek (UCLA) and Robert Calderbank (Duke).

2016–2018 **WSU Systems-on-Chip Laboratory**.
Supervised by Subhanshu Gupta (WSU).

Publications, Preprints, and Patents

2026 **Shyam Venkatasubramanian***, Sean Moushegian*, Michael Lin*, Mir Park, Ankit Singhal, and Connor Lee. Stabilizing Transformer Training Through Consensus. *arXiv:2601.22614*, 2026.

2025 Ankit Singhal*, **Shyam Venkatasubramanian***, Sean Moushegian*, Steven Strutt, Michael Lin, and Connor Lee*. Odyssey: Reconstructing Evolution Through Emergent Consensus in the Global Proteome. *bioRxiv*, 2025.

2025 **Shyam Venkatasubramanian** and Vahid Tarokh. Learn2Mix: Training Neural Networks Using Adaptive Data Integration. In *The Thirty-ninth Annual Conference on Neural Information Processing Systems*, 2025.

2025 **Shyam Venkatasubramanian***, Sean Moushegian*, Ahmed Aloui*, and Vahid Tarokh. An Information-Theoretic Lower Bound on the Generalization Error of Autoencoders. *Transactions on Machine Learning Research*, 2025. Featured Certification.

2025 **Shyam Venkatasubramanian**, Ali Pezeshki, and Vahid Tarokh. Steinmetz Neural Networks for Complex-Valued Data. In *The 28th International Conference on Artificial Intelligence and Statistics*, volume 258 of *Proceedings of Machine Learning Research*, pages 3916–3924, 2025.

2024 **Shyam Venkatasubramanian**, Bosung Kang, Ali Pezeshki, Muralidhar Rangaswamy, and Vahid Tarokh. Raspnet: A benchmark dataset for radar adaptive signal processing applications. *arXiv preprint arXiv:2406.09638*, 2024.

2024 **Shyam Venkatasubramanian***, Ahmed Aloui*, and Vahid Tarokh. Random linear projections loss for hyperplane-based optimization in neural networks. In *The Fortieth Conference on Uncertainty in Artificial Intelligence*, volume 244 of *Proceedings of Machine Learning Research*, pages 3425–3447, 2024.

2024 **Shyam Venkatasubramanian**, Sandeep Gogineni, Bosung Kang, Ali Pezeshki, Muralidhar Rangaswamy, and Vahid Tarokh. Data-driven target localization: Benchmarking gradient descent using the cramer-rao bound. *arXiv:2401.11176*, 2024.

2024 **Shyam Venkatasubramanian**, Sandeep Gogineni, Bosung Kang, Ali Pezeshki, Muralidhar Rangaswamy, and Vahid Tarokh. Data-driven target localization using adaptive radar processing and convolutional neural networks. *IET Radar, Sonar & Navigation*, volume 18, pages 1638–1651, 2024.

2023 **Shyam Venkatasubramanian**, Sandeep Gogineni, Bosung Kang, Ali Pezeshki, Muralidhar Rangaswamy, and Vahid Tarokh. Subspace perturbation analysis for data-driven radar target localization. In *2023 IEEE Radar Conference (RadarConf23)*, pages 1–5, 2023.

2022 **Shyam Venkatasubramanian**, Chayut Wongkhamthong, Mohammadreza Soltani, Bosung Kang, Sandeep Gogineni, Ali Pezeshki, Muralidhar Rangaswamy, and Vahid Tarokh. Toward data-driven stap radar. In *2022 IEEE Radar Conference (RadarConf22)*, 2022.

2021 Siyi Yang, Ahmed Hareedy, **Shyam Venkatasubramanian**, Robert Calderbank, and Lara Dolecek. Grade-ao: Towards near-optimal spatially-coupled codes with high memories. In *2021 IEEE International Symposium on Information Theory (ISIT)*, pages 587–592, 2021.

2020 Subhanshu Gupta, Erfan Ghaderi, Sudip Shekhar, **Shyam Venkatasubramanian**, and Ajith Sivadhasan Ramani. Spatial interference cancellation for simultaneous wireless and information power transfer. United States Patent US10804988B2, 2020.

Awards and Organizations

2019–Present Student Member, IEEE

2019–Present IEEE Eta Kappa Nu (HKN)

2014, 2018 DOE National Science Bowl Finalist